



Princess Sumaya University for Technology
Database Systems Project
Phase three

To You
Food Delivery Application

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Section: 3

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Introduction

We chose this topic because food delivery applications have become an essential part of our daily lives, providing convenience and efficiency for users. Additionally, this idea stood out among our options such as the smart gym concept and after discussing both ideas with Dr. Nailah, we decided that the food delivery system would be the most suitable and practical choice for our project.

how work was distributed among team members

Document was made by Abdallah Abdullah and Mohammad
Requirements was made by Abdallah Mansour edited by Abdallah Abdullah & Mohammad
ER was made by Omar and Fares edited by Abdallah Abdullah & Mohammad
EER was made by Abdallah Abdullah and Mohammad & by Omar and Fares
Mapping was made by all members

Requirements by Abdallah Mansour edited by Abdallah Abdullah & Mohammad

Food Delivery System Database entities:

Customer, Restaurant, Driver, Order, Menu_Item, Payment (Weak)

Attribute:

1) Customer (Customer_ID (PK) ,Name, Email(unique), Date_of_Birth, Address, Phone, Age) .

Composite: Address (Street , city , building_No , floor) , Multivalued: Phone , Derived:Age from Date_of_Birth).

2) Restaurant(Restaurant_ID (PK) Name(Unique), Location, Rating).

3) Driver(Driver_ID (PK), Name, License_No(Unique) , Hire_Date, Years_of_Service).

#Derived: Years_of_Service (from Hire_Date).

4) Order(Order_ID (PK), Order_Time, Delivery_Time, Status).

5) Menu_Item(Item_ID (PK) , Name, Description, Price).

6) Payment (Weak) (Payment_No (Partial Key), Amount, Payment_Date).

Weak Entity: depends on Order.

Relationship:

R1: Customer place one or more order ,order must be placed by one customer only .

R2: Recursive relationship is the relationship where in which the system tracks customer referrals, and in this relationship a customer may refer more than one customer, but a customer is referred by a customer at max.

R3: (Weak Relationship): Order identifies Payment. a Payment cannot exist without associated order .

R4 : Restaurant Offers menu_items , every menu_item must be offered by exactly one restaurant.

R5: Specialization (Disjoint) The Menu_Item, entity has two specializations into two disjoint subclasses that are based on the type:

1) Main_course

2) dessert

It's a disjoint and a specialization which means every Menu_item is either a main_course or a dessert.

R6: an order can contain multiple Menu_items, and a Menu_Item can be in many Orders.

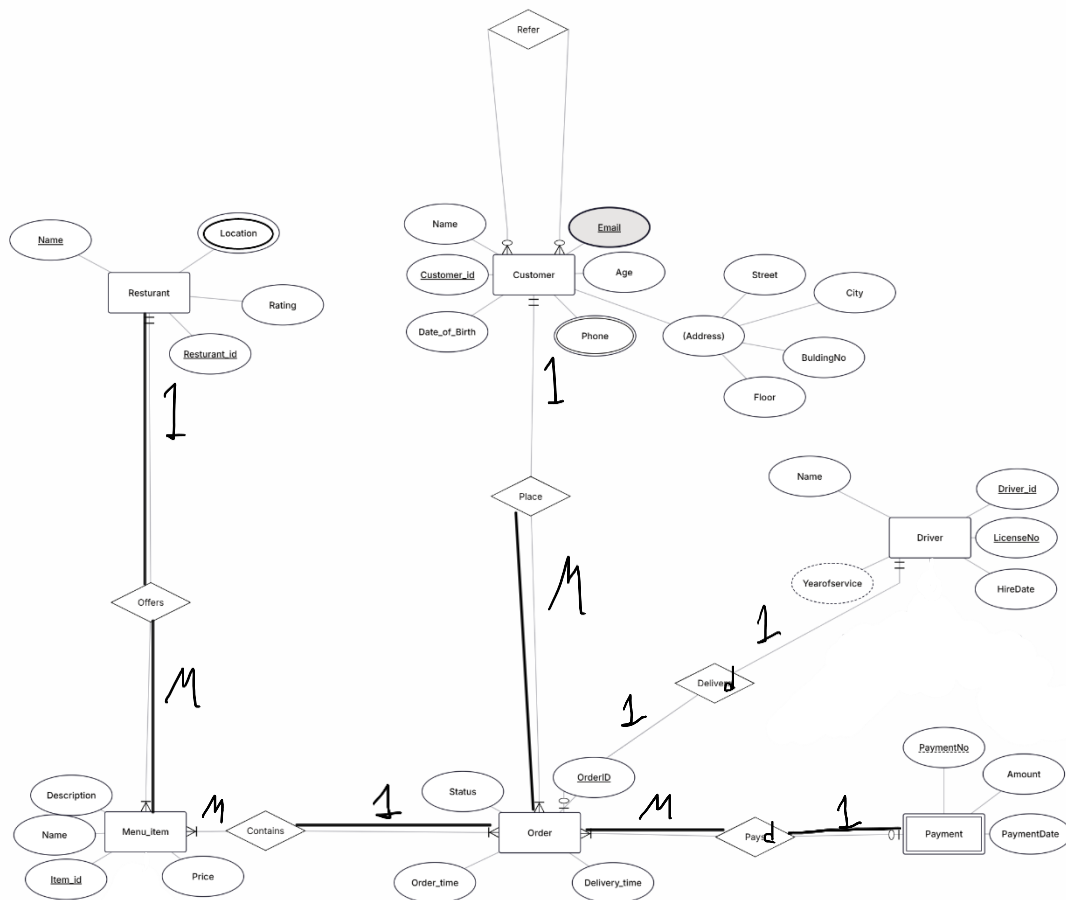
R7: A driver delivers one order 1:1 relationship , not every driver is actively delivering , order must be taken from the driver .

R8: Specialization 2 (Overlapping): The driver entity is specialized based on the vehicle type:

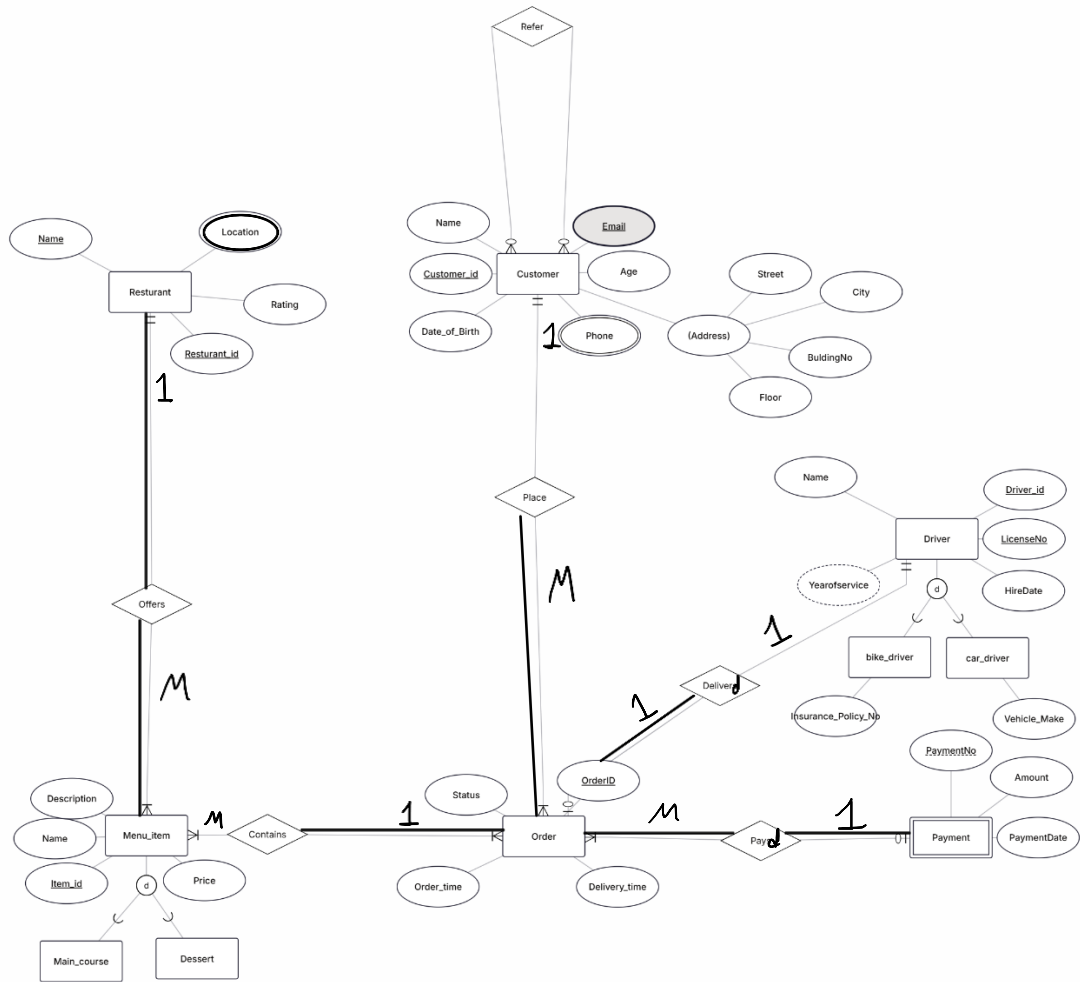
- 1) bike_driver (Additional attribute: Insurance_Policy_No)
- 2) car_driver (Additional attribute: Vehicle_Make)

This specialization is overlapping and partial specialization which allows a driver to be associated with multiple vehicle types or non.

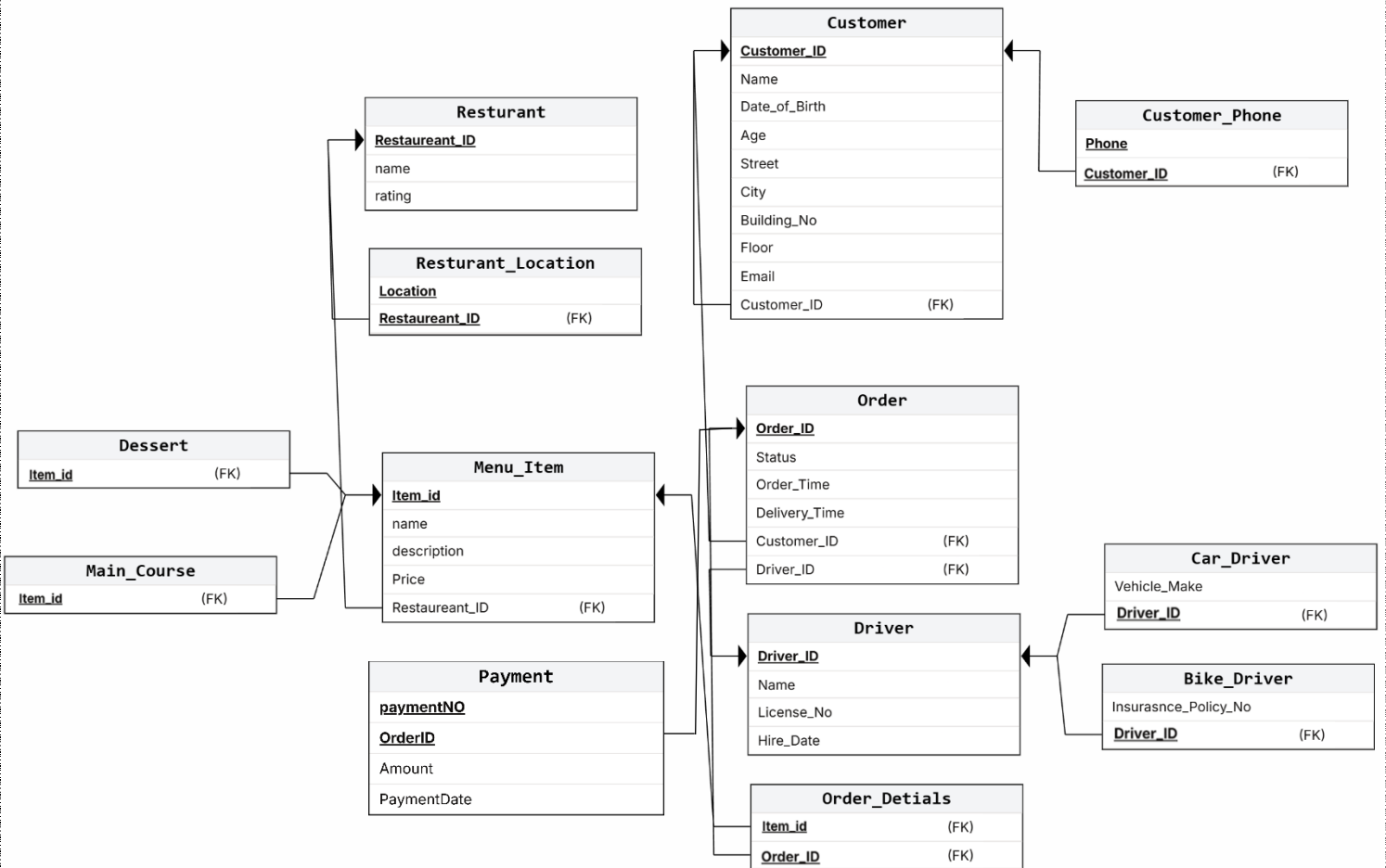
ER by Omar and Fares edited by Abdallah Abdullah & Mohammad



EER by Abdallah Abdullah & Mohammad & Omar and Fares



Mapping by all team members



1&2 by Abdallah Abdullah

CUSTOMER_ID	NAME	DATE_OF_BIRT	AGE	STREET	CITY	BUILDING_NO	FLOOR	EMAIL	REFERRER_ID
1	ahmad	15-FEB-95	31	king abdullah	Amman	12	3	ahmad@gmail.com	
2	abdullah mansour	10-JUN-90	35	wasfi altal	Amman	5	2	abdman@gmail.com	1
3	abdallah mohammad	20-SEP-00	25	Khalil elsakit	Amman	18	1	abdmoh@gmail.com	2
4	mohammad shahwan	05-DEC-98	27	shawerma st	Amman	22	4	mhmshah@gmail.com	
5	omar	30-MAR-92	33	jordan st	Amman	9	1	omarzaga@gmail.com	3

PHONE	CUSTOMER_ID
0770750749	3
0778457199	4
0791190345	5
0791234567	1
0792370283	2

RESTAURANT_ID	NAME	RATING
1	Almousili	4.5
2	Sushi Crush	4.8
3	fire fly	4.2
4	buffalo	4.8
5	Sweet	4.7

LOCATION	RESTAURANT_ID
Abdali	5
Jabal Amman	3
Mecca St	4
abdon	2
almanhal circule	1

DRIVER_ID	NAME	LICENSE_NO	HIRE_DATE
1	faris smarh	L12345	01-JAN-20
2	abdalrazaq	L67890	15-MAR-19
3	zaid ahmad	L54321	10-MAY-21
4	nader ali	L98765	22-JUL-18
5	batman	L14323	01-SEP-22
6	ali mhmd	L52345	01-JAN-20
7	tariq	L67790	15-MAR-19
8	nour	L54221	10-MAY-21
9	sarah	L92765	22-JUL-18
10	leen	L11223	01-SEP-22

DRIVER_ID VEHICLE_MAKE

1 Ford fusion
2 Honda Civic
3 Ford Focus
4 corola
5 camrey

DRIVER_ID INSURANCE_POLIC

6 INS12345
7 INS67890
8 INS54321
9 INS98765
10 INS11223

ORDER_ID STATUS

ORDER_TIM

DELIVERY_

CUSTOMER_ID

DRIVER_ID

1 Delivered
2 Pending
3 Cancelled
4 Delivered
5 Delivered

01-JAN-26 01-JAN-26
02-JAN-26
03-JAN-26
04-JAN-26 04-JAN-26
05-JAN-26 05-JAN-26

1 1
2 2
3 9
4 4
5 7

PAYMENTNO

ORDERID

AMOUNT

PAYMENTDA

1 1 20.5 01-JAN-26
2 2 35.75 02-JAN-26
3 3 15 03-JAN-26
4 4 40.25 04-JAN-26
5 5 10 05-JAN-26

ITEM_ID	NAME	DESCRIPTION	PRICE
RESTAURENT_ID			
1	shawarma double	shawarma double and fries and pepsi	4.5
2	California Roll	8 pcs sushi roll	8.75
3	Cheeseburger	Beef burger with cheese	10
4	Spaghetti	Classic spaghetti with buffalo sauce	9.5
5	Chocolate Cake	chocolate dessert with cream on top	6
6	Margherita Pizza	Classic Margherita with mozzarella and basil	11
7	Tuna Sushi Roll	8 pcs tuna sushi roll	9.5
8	BBQ Chicken Burger	Grilled chicken with BBQ sauce	11.5
9	Fettuccine Alfredo	Creamy fettuccine with parmesan	10
10	Cheesecake	Classic New York style cheesecake	7
11	Veggie Pizza	Pizza with bell peppers, olives, and mushrooms	10.5
12	Salmon Sushi Roll	8 pcs salmon sushi roll	10.25
13	Double Cheeseburger	Two beef patties with cheese	12
14	Lasagna	Layers of pasta with meat and cheese	11.25
15	Brownie	Chocolate brownie with ice cream	6.5

ITEM_ID	NAME	DESCRIPTION	PRICE
RESTAURENT_ID			
23	Creme Brulee	Rich custard with caramelized sugar	7.25
24	Chocolate Mousse	Fluffy chocolate mousse	6.5
25	Banana Split	Banana with ice cream and chocolate syrup	6.75

ITEM_ID
1
2
3
4
6
7
8
9
11
12
13
14

ITEM_ID
5
10
15
23
24

3 by Mohammad & Abdallah Abdullah

4 by Fares

5 by Abdallah mansour

6 by omar